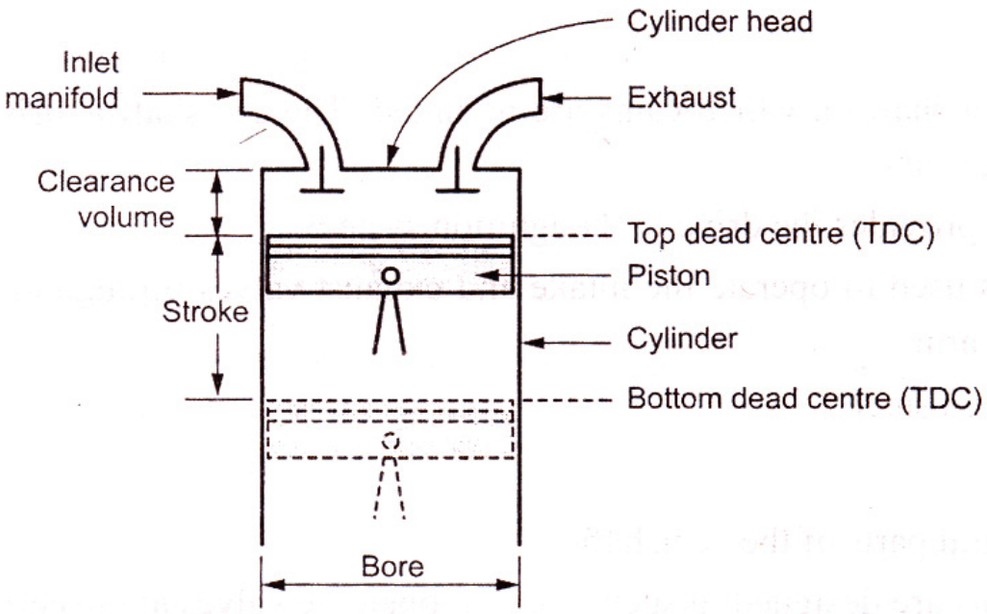


Figure 4.3 Technical Specification of the Engine

9. Compression ratio is the ratio of total volume of the cylinder to clearance volume.
10. Indicated power is defined as the power developed by combustion of fuel inside the engine cylinder.

$$IP = \frac{P_m LAN}{60 \times 1000}$$

I.P. = Indicated Power, kW

P_m = Mean Effective Pressure, N/m^2

L = Stroke Length, m

A = Cross section area of Piston, m^2

N = Crankshaft RPM (for 2 Stroke engine N, for 4 Stroke engine N/2)

11. Brake power is the actual work output of an engine or the actual work available at the crankshaft. It can be measured with the help of brake dynamometer.

$$B.P = \frac{2\pi NT}{60 \times 1000}$$

B.P. = Brake Power, kW

N = Crankshaft RPM

T = Torque or Resisting torque in the dynamometer, Nm and

12. Frictional power: Engine brake power is always less than Indicated power, due to frictional losses at the working surfaces like bearings, piston rings and valves. The power loss due to friction is called as frictional power.
13. Friction power, F.P. = Indicated power, I.P. – Brake power, B.P.

Efficiency: The ratio of power output and power input to the engine is called as Efficiency. It is calculated by Volumetric Efficiency, Thermal Efficiency and Mechanical Efficiency.

Volumetric Efficiency is defined as the ratio of the actual volume of air inducted during the intake stroke to the theoretical volume of a cylinder.

Thermal Efficiency is the ratio of the useful work obtained to the heat supplied to the engine.

Mechanical Efficiency is defined as the ratio of flywheel output to the useful work obtained from the engine.

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